What is Anomaly Detection?

What is **Anomaly Detection**?

In **data analysis**,

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In **data analysis**, **anomaly detection**

What is **Anomaly Detection**?

In **data analysis**, **anomaly detection** is the **identification of rare** items, events or observations

What is **Anomaly Detection**?

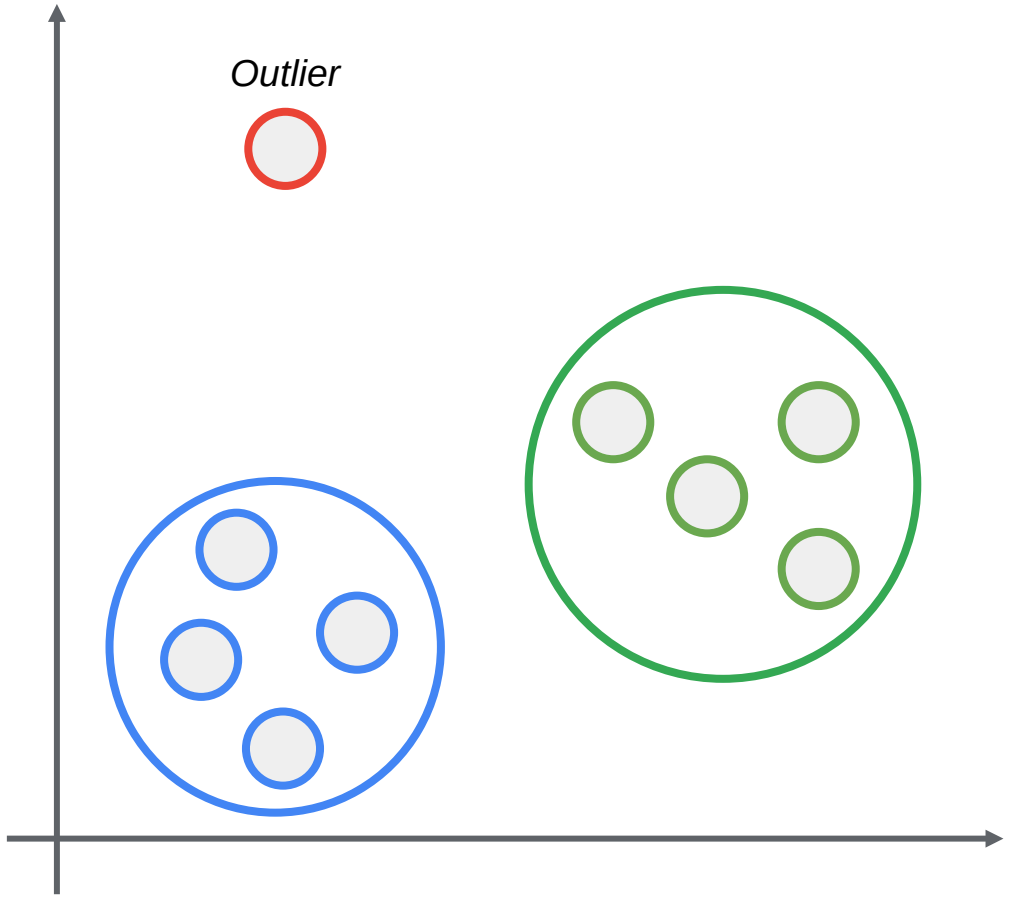
In **data analysis**, **anomaly detection** is the **identification of rare** items, events or observations which **raise suspicions**

What is **Anomaly Detection**?

In **data analysis**, **anomaly detection** is the **identification of rare** items, events or observations which **raise suspicions** because they **differing significantly**

What is **Anomaly Detection**?

In **data analysis**, **anomaly detection** is the **identification of rare** items, events or observations which **raise suspicions** because they **differing significantly** from the **majority of the data**.


Outlier

Multiple **Application Domains**



Health



Industry



Security

Multiple **Sensors**



Health

ECG Sensor



Industry

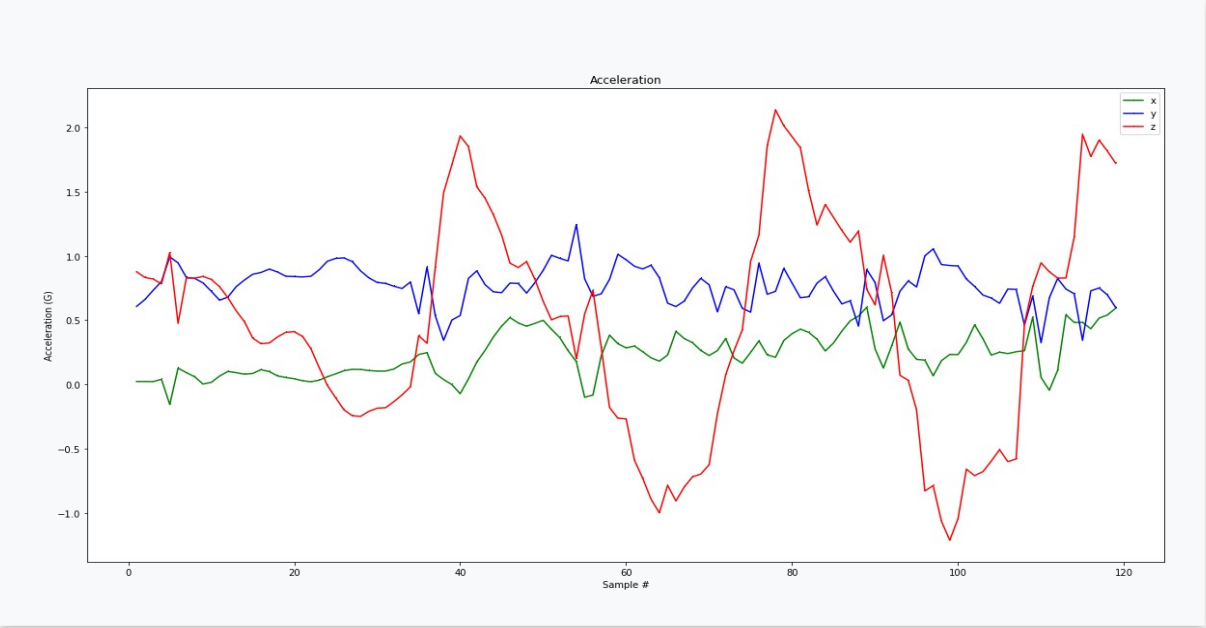
Accelerometer



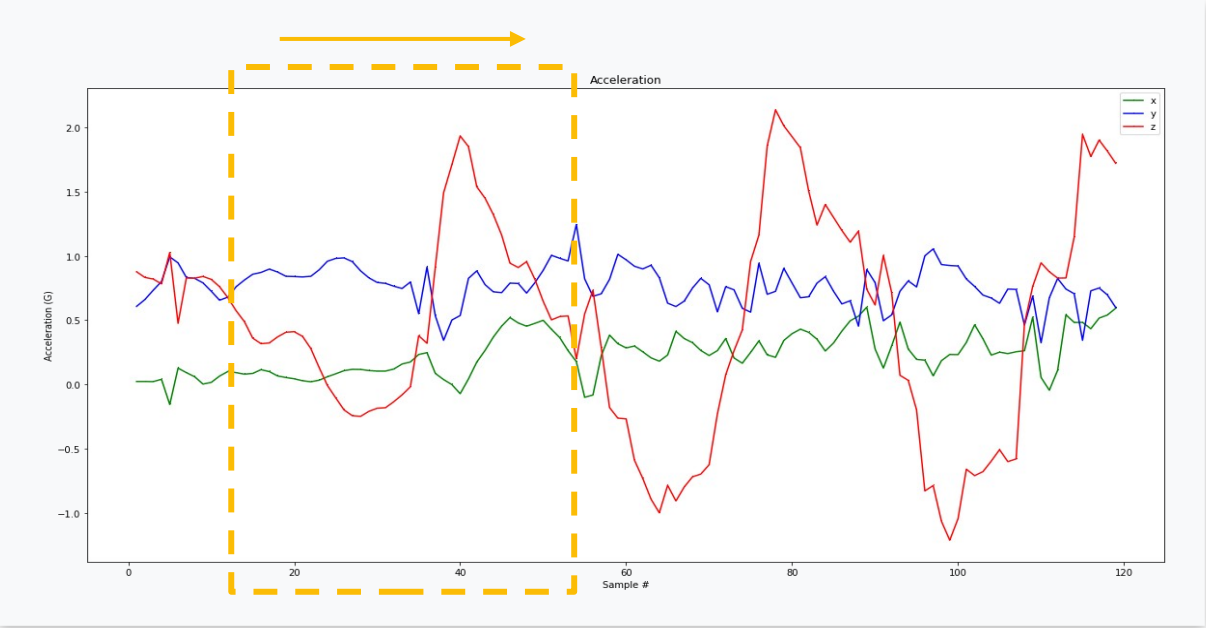
Security

IR Motion Sensor

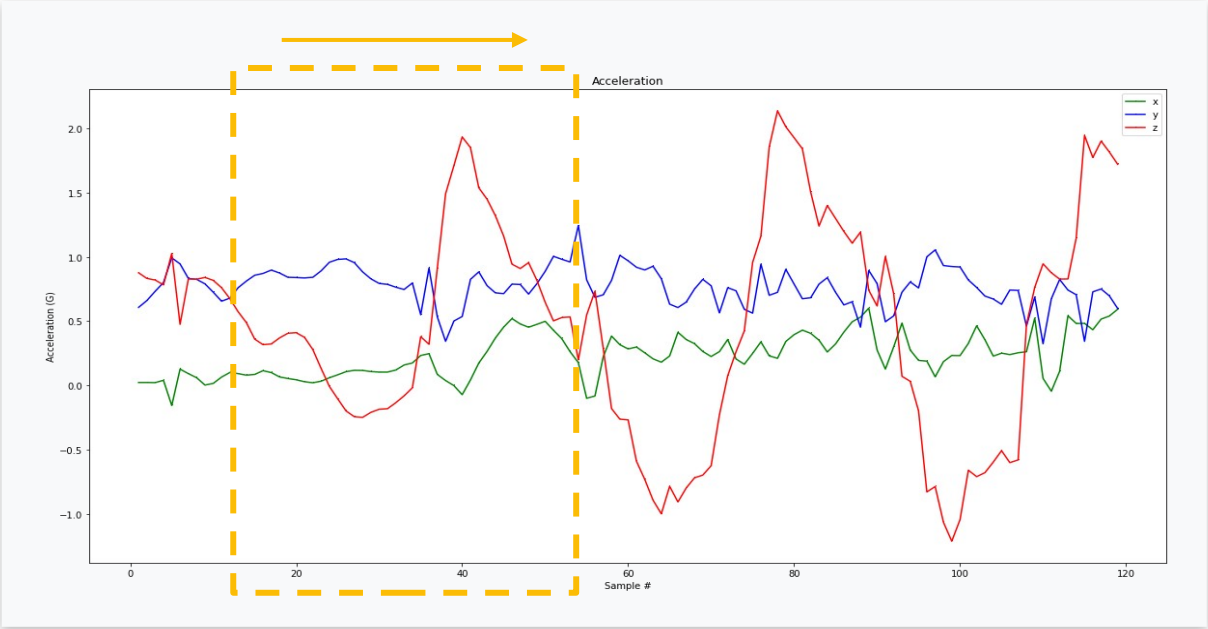
Multiple **Sensors**



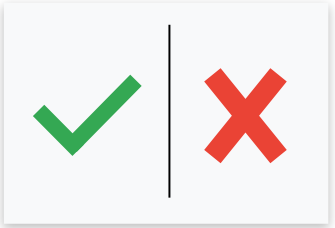
Multiple **Sensors**



Multiple **Sensors**

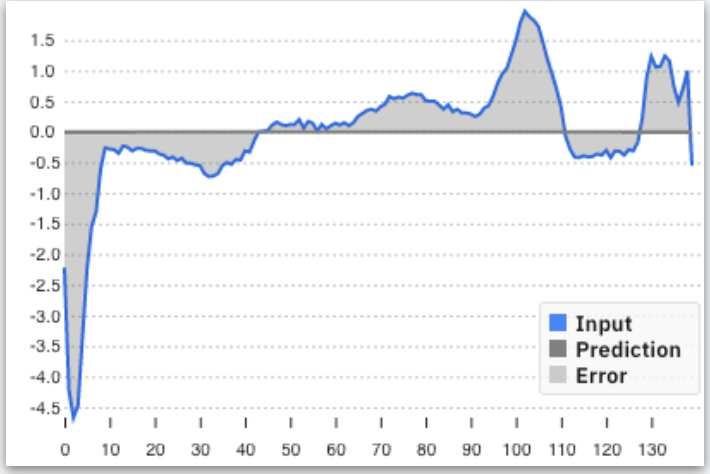


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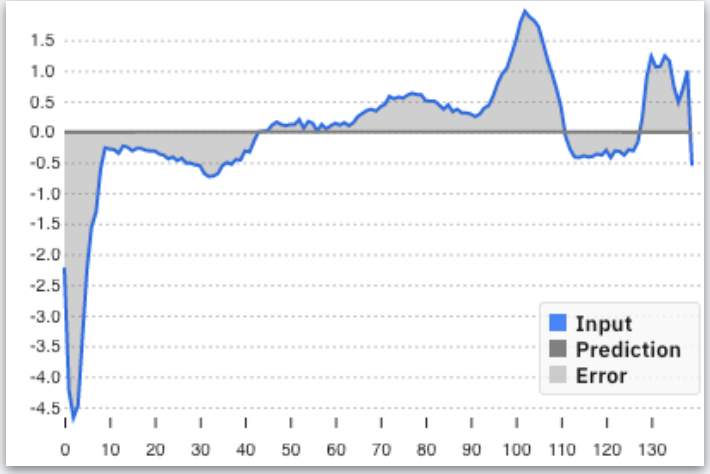
Three Fundamental Aspects of AD

1. **Input**



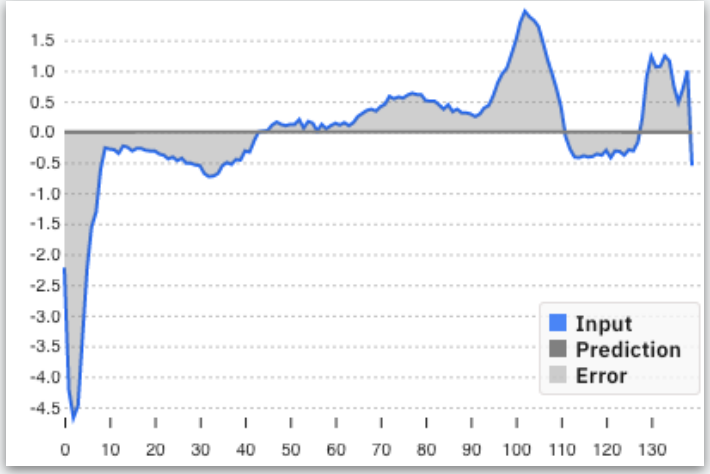
Three Fundamental Aspects of AD

1. Input
2. **Prediction**

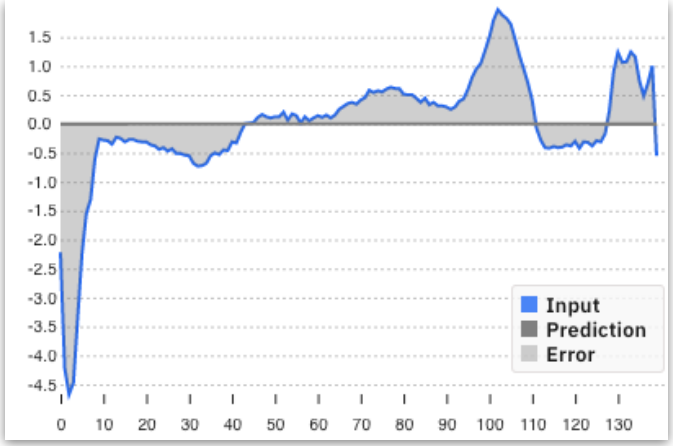


Three Fundamental Aspects of AD

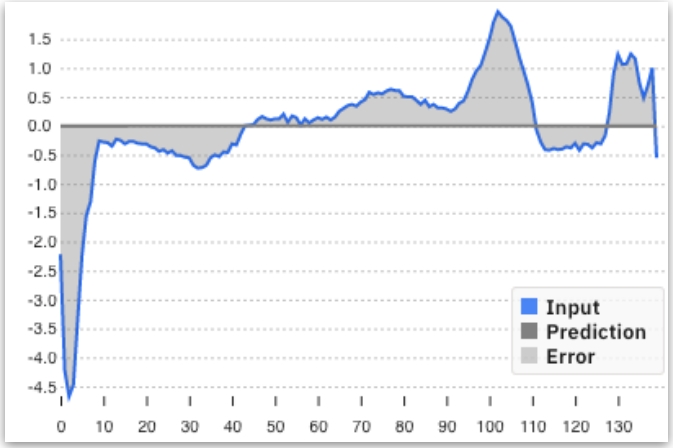
1. Input
2. Prediction
3. **Error**



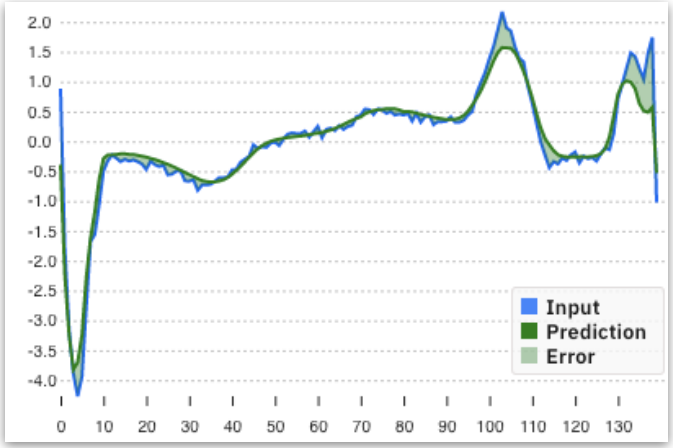
Poor Prediction



Poor Prediction



Good Prediction



Why **TinyML**?

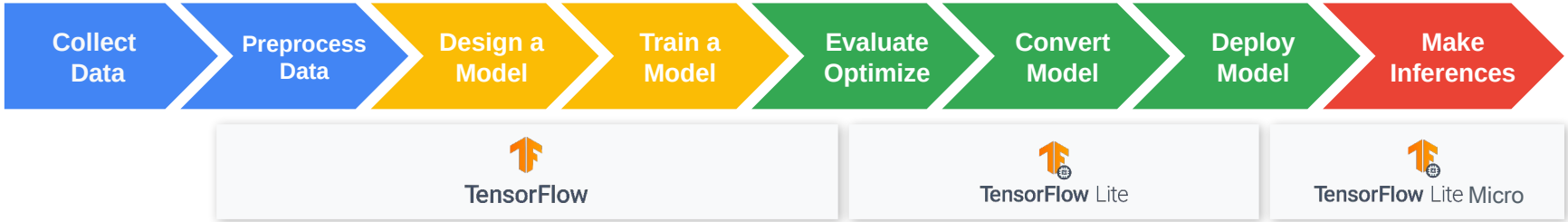




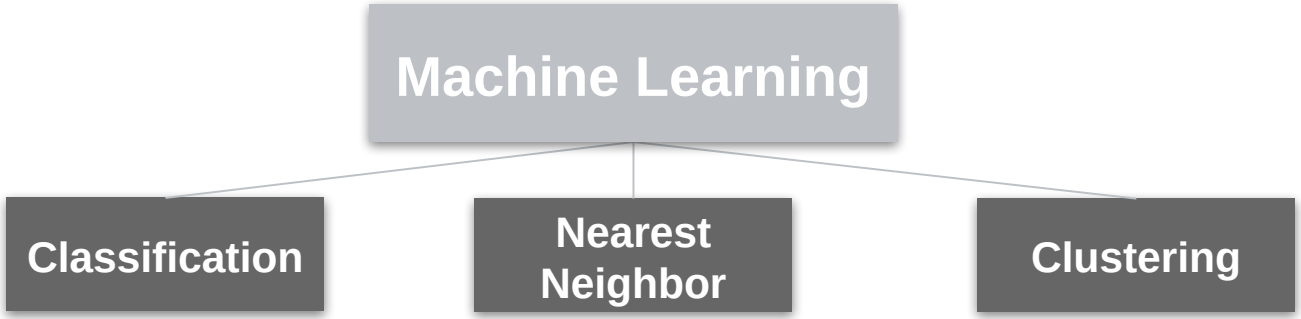


Machine Learning

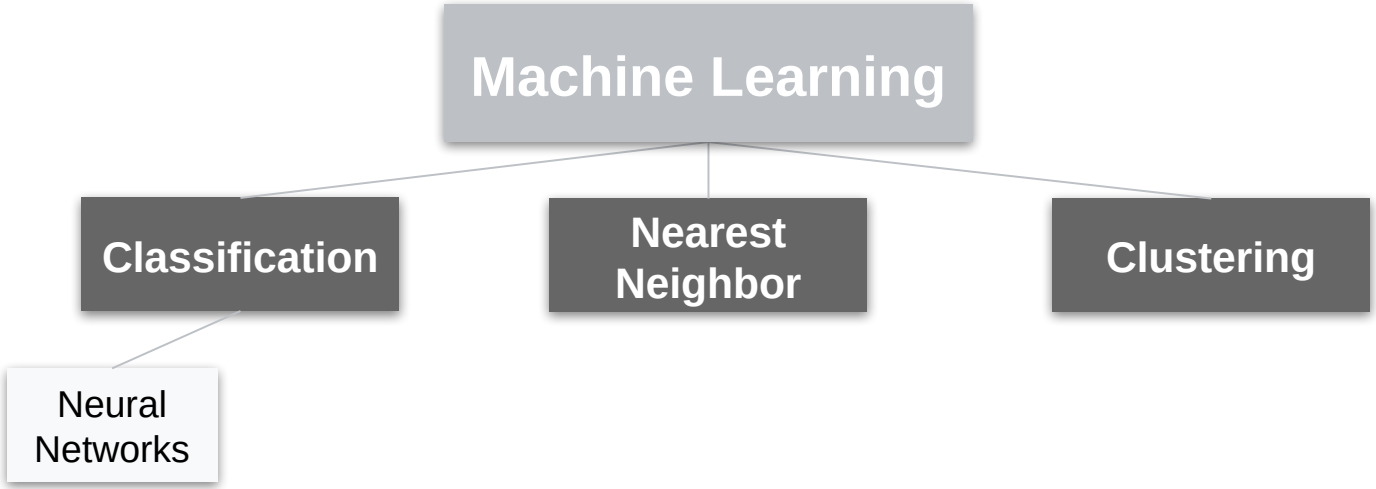
Machine Learning



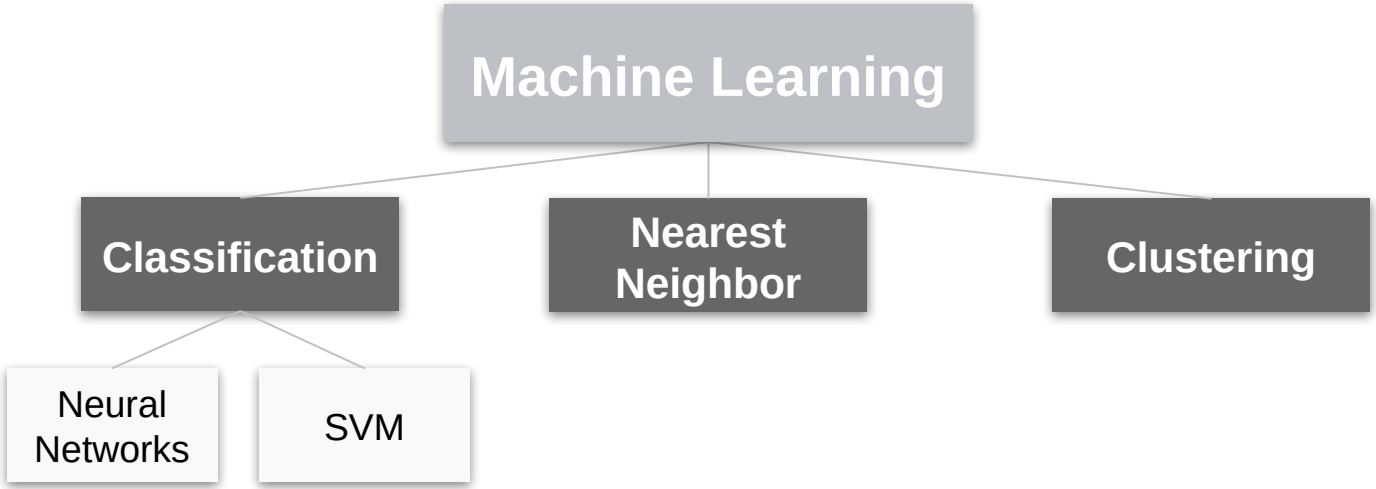
It’s **not all** deep learning



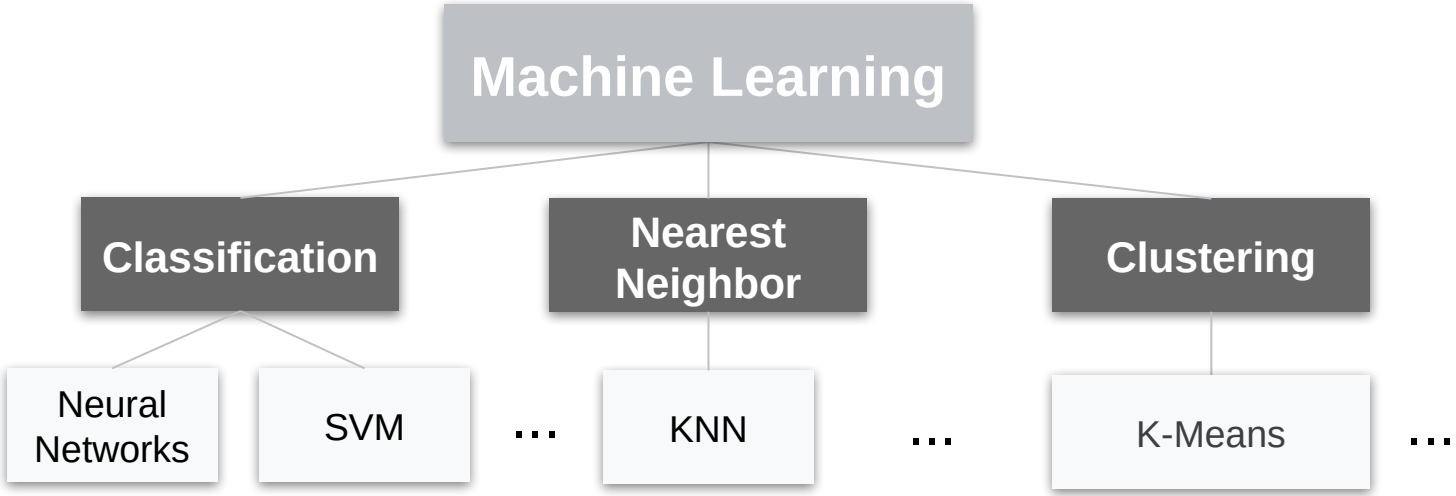
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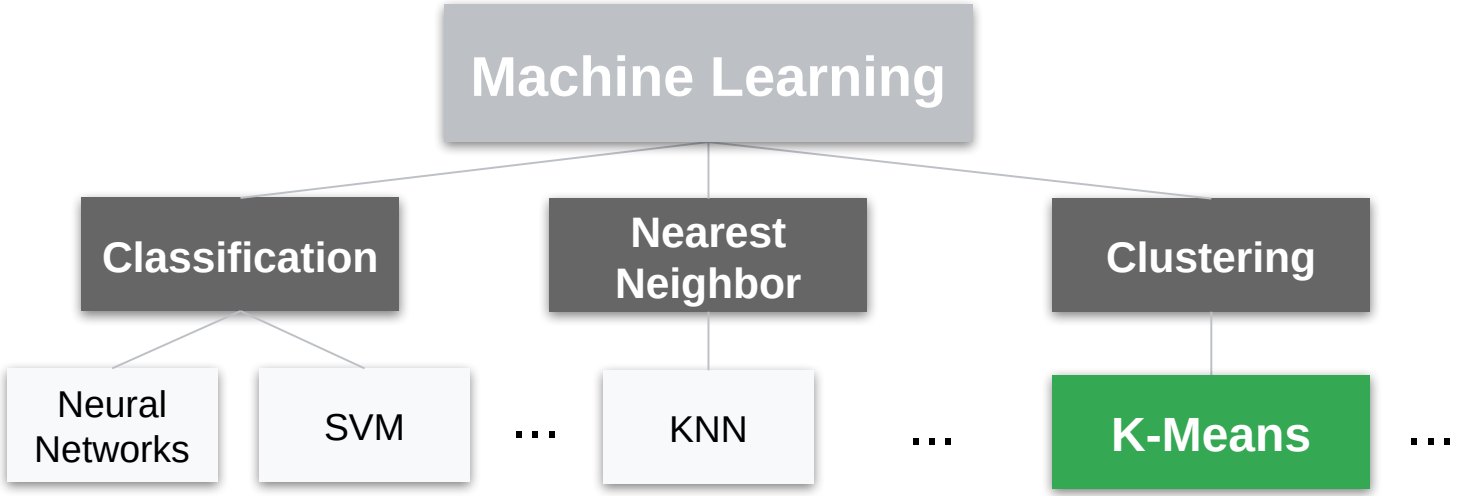
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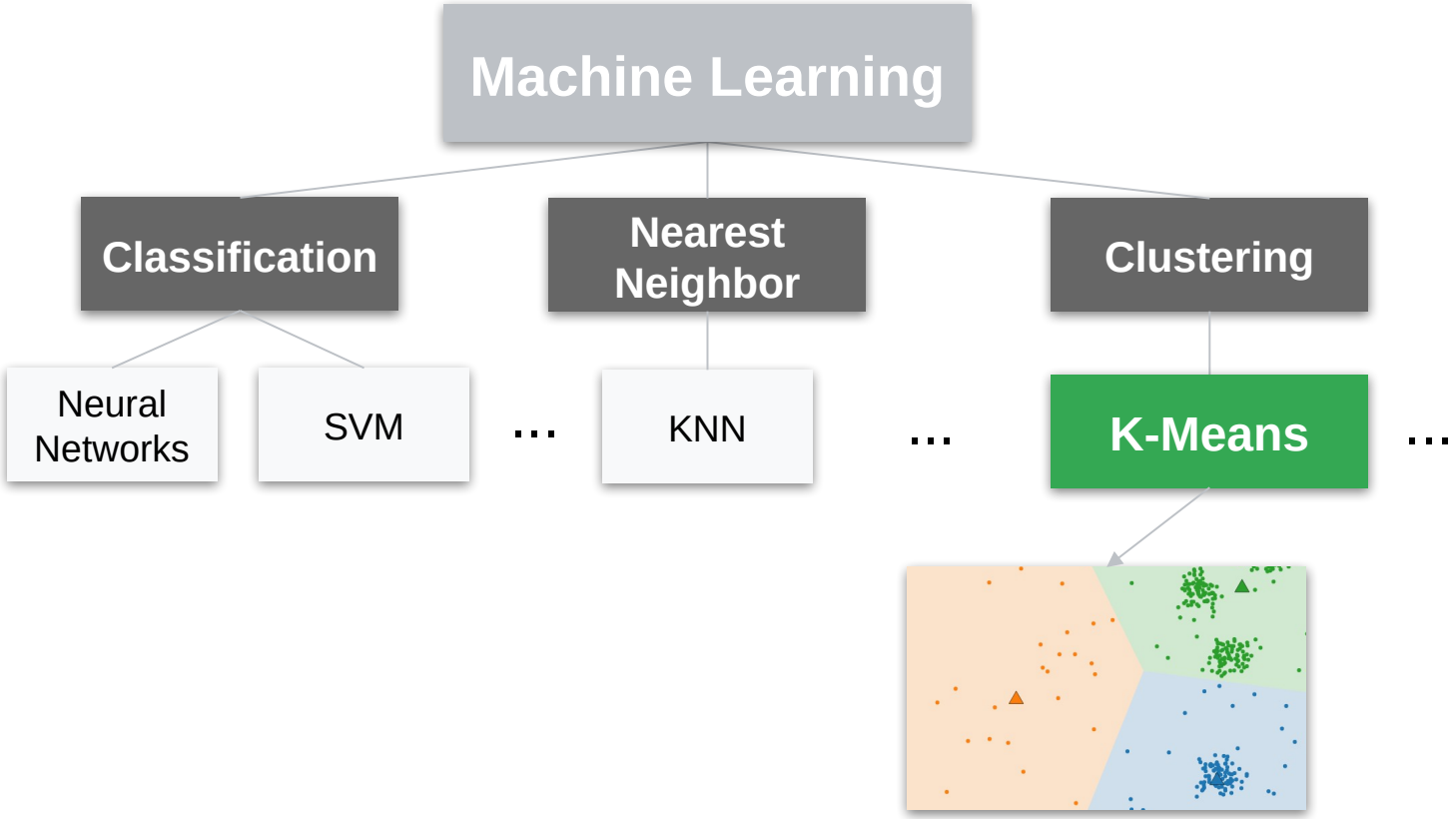
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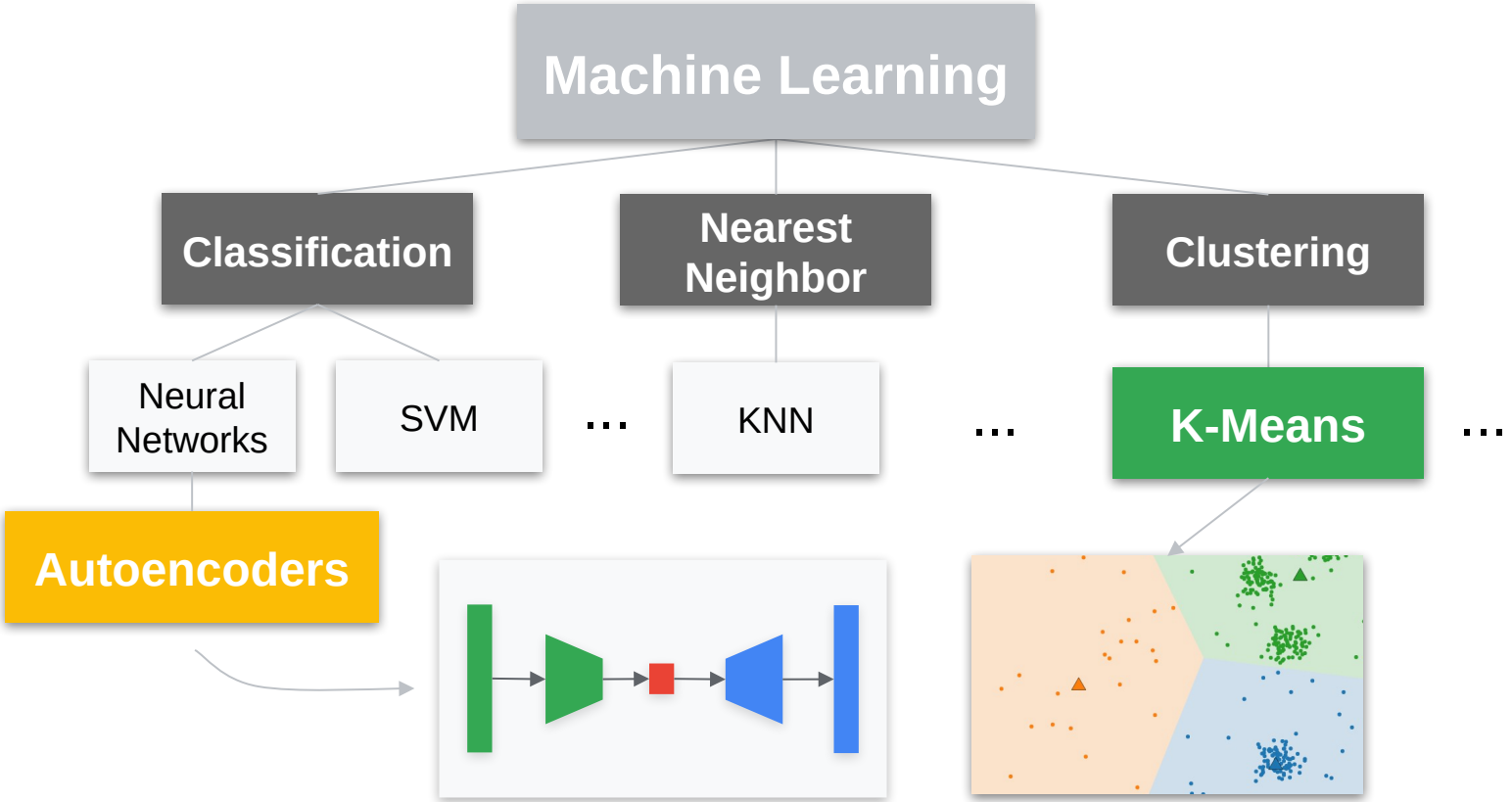
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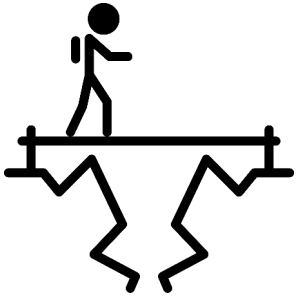
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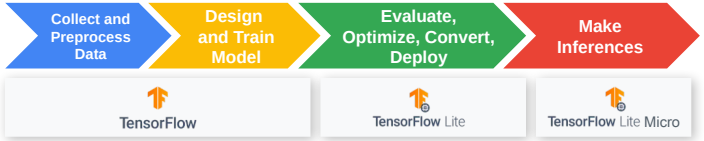
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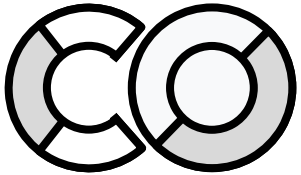
What are we going to learn?



Challenges with an Anomaly Detection Application



Anomaly Detection **ML Pipeline**



Training, Testing in Colab